mal die size, power consumption and resource configuration required for their end application. They define quantity of LUTs, embedded memory blocks, DSP blocks, Speedcore aspect ratio and IO port connections. They can make tradeoffs between power and performance.

Achronix delivers GDS II of Speedcore IP to integrate directly into an SoC, and a custom, full-featured version of ACE design tools to design, verify and program the functionality of Speedcore eFPGA. Designers have long sought the inherent advantages that could come from embedding FPGA functionality as IP into SoCs for a host of different high-performance applications.

Achronix has now delivered eFPGA IP to customers who are developing high-performance computing products, where offloading compute-intensive functions from processors to FPGA IPs can provide a tremendous performance boost. While an exciting opportunity for Achronix, seeing FPGA IP become a reality is great news for the semiconductor industry, particularly considering the large market opportunity for compute high-performance applications.

Speedcore eFPGA is an embeddable IP. It does not include programmable IO, and is designed to be surrounded by end-user ASIC.

***Speedcore is the optimal hardware accelerator.*** FPGAs are the optimal hardware accelerator solution, because accelerators need to be updated with new functionalities as algorithms are constantly changing. Standalone FPGAs are a convenient and practical solution for low- to medium-volume applications. Whereas, Speedcore is the optimal solution for high-volume applications and offers significant advantages, as given below.

*Lower power.* Speedcore has direct wire connections to the SoC, which eliminate large programmable IO buffers found in standalone FPGAs. Programmable IO circuitry accounts for half of the total power consumption of standalone FPGAs.

Speedcore is sized exactly to the requirements of the customer's end application. The customer can tune the process technology to tradeoff performance for lower power.

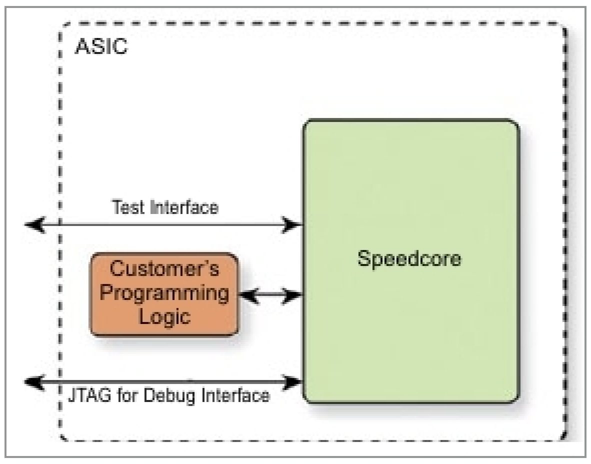


*Fig. 4: Embedded Speedcore*

*Higher interface performance.* Speedcore offers dramatically higher interface performance than standalone FPGAs in the form of lower latency. It is connected to the ASIC through an ultra-wide parallel interface, whereas standalone FPGAs typically connect through a high-latency serialiser/deserialiser (SerDes) structure.

*Lower system cost.* Speedcore die size is much smaller than standalone FPGAs, because the programmable IO buffer structure is eliminated.

FPGAs have high pin counts, which dictate the PCB layer count to support FPGA BGA package escape routing.

Additionally, Speedcore eliminates the need for all support components around the standalone FPGA, including power regulators, clock generators, level shifters, passive components and FPGA cooling.

*Higher system reliability and yields.* Integrating FPGA functionality into an ASIC improves system-level signal integrity, and eliminates reliability and yield loss associated with a standalone FPGA on the PCB.

***Process technology.*** Speedcore is architected in modular fashion to support flexibility for customers to define their resource requirements, and for Achronix to rapidly configure Speedcore IP for delivery.

Additionally, the modular architecture allows Achronix to easily port the technology to different process technologies and metal stacks. Speedcore is available on TSMC 16FF+, and is in development on TSMC 7nm.

***Evaluating Speedcore.*** Achronix's ACE design tools include an example of Speedcore, where customers can compile their designs to evaluate Speedcore quality-of-results for performance, resource usage and compilation times.

Additionally, Achronix has complete documentation on Speedcore functionality and ASIC integration methodologies.

***Physical integration with customer ASICs.*** Speedcore eFPGA is provided as a fixed-transistor-layout building block that integrates with industry-standard ASIC flows, such as Synopsys Design Compiler and IC Compiler. The following collateral is provided:

1. Verilog definition of logical connectivity at boundary
2. Liberty timing library for timing closure at boundary
3. LEF defining physical floorplan, pins and metal blockages
4. GDS/Oasis physical database

## Xilinx FPGA

Xilinx has advanced architecture features like enhanced DSP slices incorporating 27x18-bit multipliers and dual adders that enable a massive jump in fixed- and IEEE standard 754 floating-point arithmetic performance and efficiency.

In addition, it has step-function increase in 3D IC inter-die bandwidth for virtual monolithic design, massive IO bandwidth and dramatic latency reduction through multiple integrated ASIC-class blocks for 100G Ethernet with RS-FEC, 150G Interlaken and PCIe generation 4, and static- and dynamic-power gating across a wide range of functional elements, yielding significant power savings.

Next-generation security with advanced approaches to AES bitstream decryption and authentication, key-obfuscation and secure device programming, and DDR4 support up to 2666Mbps for massive memory interface bandwidth are additional features. Ultra-RAM provides massive on-chip memory for SRAM device integration, innovative IP interconnect optimisation technology for an additional 20 to 30 per cent advantage in performance/per watt, MPSoC technology, combining soft and hard engines for real-time control, graphics and video processing, waveform and packet processing, and multi-level security, safety and reliability, and more.